NMIMS MPSTME · B.TECH COMPUTER ENGINEERING CAPSTONE · V5.0

AEGIS-CARE

Delete the bad memory and the damage stays. AEGIS-Care rebuilds everything it touched.

0.000 RESIDUAL HARM LOWER IS BETTER

1.000 DESCENDANT RECALL & PRECISION

1.000 BENIGN STATE RETAINED

RESEARCH BOUNDARY FULLY SYNTHETIC FHIR · MECHANISM EVIDENCE, NOT A DIAGNOSIS, CLINICAL VALIDATION OR PATIENT-BENEFIT CLAIM

01 The incident

A corrupted patient alias is summarised, cached and handed over. Deleting the original entry does not revoke the **downstream clinical memories** already in use.

FIG. 1 — PROPAGATION

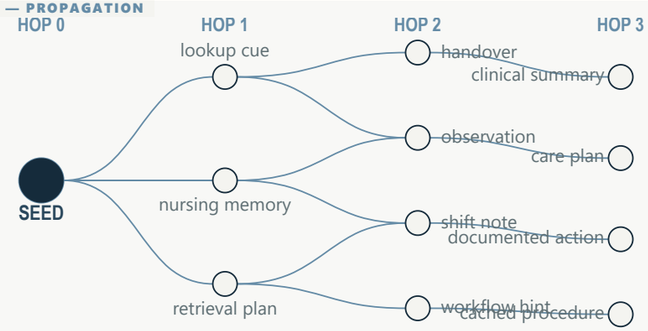
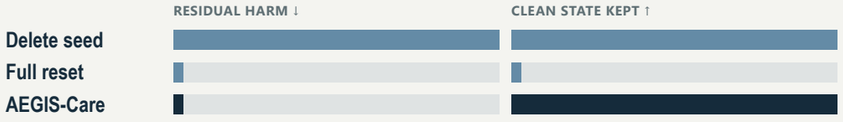


FIG. 2 — WHAT EACH RESPONSE COSTS



Why it matters: the recovery target is not "delete a row." It is to restore every affected future decision while preserving unrelated clinical memory and the privacy boundary.

OBJECTIVES

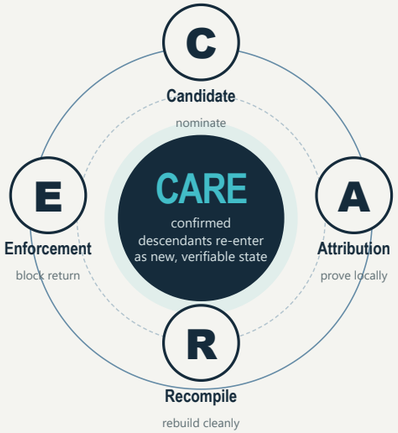
Discover
recover descendants even when provenance edges are missing.

Repair safely
rebuild only causally affected memories from trusted FHIR.

Prove closure
preserve clean state, privacy and deletion integrity.

02 The CARE loop

Proposed solution / technical design: each runtime joins a bounded recovery protocol. The coordinator receives commitments and decisions—not the clinical prose itself.



KEY FEATURES / INNOVATION

FIND BEYOND LINEAGE
Exact edges plus receiver-scoped semantic sketches and FHIR support fingerprints.

REQUIRE CAUSAL PROOF
Local counterfactual replay asks whether the patient or action actually changes.

REBUILD FROM SOURCE
Confirmed descendants are reconstructed from canonical FHIR or held for review.

MAKE DELETION STICK
Signed tombstones, certificates and a resurrection firewall close the incident.

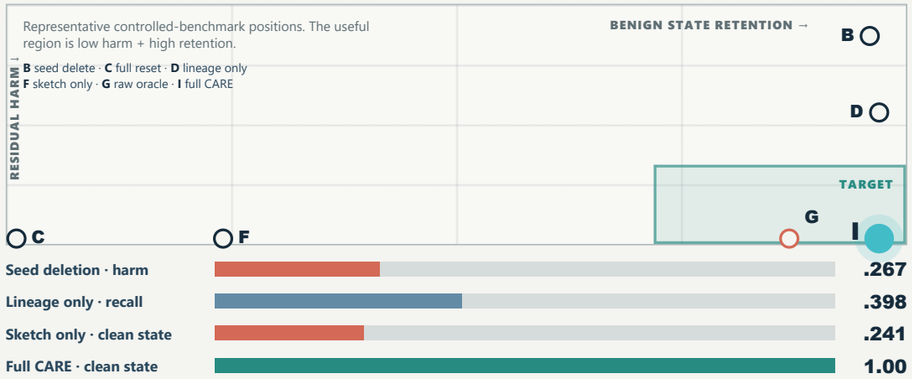
Privacy design: raw content stays with its owning role. Capsule fields are purpose- and recipient-scoped; recovery decisions are auditable and independently verifiable.

03 Prototype & evidence

Current progress: a runnable end-to-end prototype, role-aware UI, evidence verifier and paired benchmark suite. Full CARE is the only non-oracle condition that reaches every target together.

CANONICAL CONTROLLED CAMPAIGN
900 runs · 100 incidents
4 causal families × 3 depths × 5 provenance regimes × 9 recovery conditions.

EXTERNAL-FORMAT CAMPAIGN
108 runs · 12 incidents
10 public Synthea patients · 3,623 FHIR resources · 7,003 rewritten references.



OBSERVED RESULT · BOTH CAMPAIGNS
0 harm · 100% recall · 100% precision
With all clean state retained, zero unauthorised exposure and zero deletion resurrection.

Evidence is reproducible.
Deterministic experiments, SHA-256-bound manifests, signed recovery certificates and 225 repository tests support mechanism claims.

Not evidence of: patient benefit, hospital deployment, certification or clinical effectiveness.

STUDENT DETAILS & GUIDANCE

Lakshit Sachdeva · SAP: **TBC**
Trayee Nayampalli · SAP: **TBC**
Anannya Samudra · SAP: **TBC**
Sohum Sawant · SAP: **TBC**
Project Mentor: **To Be Confirmed**

ACADEMIC & TECHNICAL CONTEXT

SVKM's NMIMS · MPSTME
Department of Computer Engineering · B.Tech Capstone
Project: AEGIS-Care — Privacy-Bounded Memory Recompiler
Built with: Python, FastAPI, FHIR-shaped data, SQLite, SHA-256 and Ed25519.

SOCIAL / ENVIRONMENTAL IMPACT

Social: aims to reduce the persistence of wrong-patient and unauthorised clinical-agent memory while retaining useful, unrelated context and respecting role boundaries.
Environmental: targeted repair may avoid repeated full-memory resets and recomputation; energy savings have **not yet been measured**.

FUTURE SCOPE / EXPECTED OUTCOME

Clinician-supervised usability study · governed FHIR sandbox integration · stronger privacy-preserving fingerprints · calibrated thresholds · larger external synthetic cohorts.
Expected outcome: independently verifiable, selective recovery without raw-content centralisation.
Synthetic research prototype · not clinically validated.